ChronicCare AI — Doctor Review Pack

11 short decisions for clinical review · November 2026 · FAST-NUCES Islamabad final-year project (Usman Bari, Hassaan Mudassar, Muhammad Saad)

What this is. ChronicCare AI is a student prototype that helps adults with type 2 diabetes and/or high blood pressure do a short daily check-in (in English, Urdu or Roman Urdu) and routes concerning answers to a doctor. It never diagnoses, never gives medicine doses, and never lowers a level that the safety rules set. No real patients use it; all testing is on made-up patients.

What we need from you. For each item: tick one option (or write your own), and add any comment. It's fine to skip an item you'd rather not decide. Then fill the last page. A photo of the ticked pages on WhatsApp is enough.

Levels the app uses: Emergency (call now) · **Urgent** (same-day clinical contact) · **Review** (a doctor looks at it) · **Routine** (nothing extra).

Reviewing doctors

- Dr Shaiza Jabbar, Medical Officer, Hearts International Hospital
- Captain Dr Talha Yasin, CMH Rawalpindi
- Dr Anam Zafar

Items 1 and 2 matter most for patient safety. If you only have a few minutes, please answer those two.

1 Danger signs typed as free text (Urdu, Roman Urdu, English)

WHAT THE APP DOES TODAY

The structured questions (chest pain? weakness on one side? etc.) and the reading-based rules work. But when a patient *types* a danger sign in their own words, the app only recognises fixed phrases. In tests with new messages written by team members:

- 20 new Urdu-script danger messages: the current app recognised **3 of 20**; an expanded phrase list recognised **7 of 20**. False alarms: 2 of 20 safe messages (both about past events, e.g. "chest pain last year, fine now").
- 15 new mixed-language danger messages: the version patients use today recognised **1 of 15**.

WHAT WE PROPOSE

Recognise **groups of words** instead of fixed phrases, in all three languages: body part + severity or sudden onset (e.g. chest + pressure/heaviness); "one side" + weakness/numbness; breathing difficulty; fainting / not responding; vomiting blood or black stool; worst or sudden severe headache; sudden loss of vision. With checks for "no / not / last year / now fine".

A match can only **raise** the check-in to **Review** and show the line *"If this is an emergency, call 1122."* It never lowers a level, and on its own it does not declare an emergency. It stays switched off until you approve it.

YOUR DECISION

☐ Approve the word-group approach (raise to Review + "call 1122" line)

☐ Approve, but these groups should go straight to **Emergency**:

_____

☐ Don't approve / other: _____

Is "Rescue 1122" the right number to show nationwide? Yes ☐ No ☐ Better:

_____

2 Glucometer shows "LO" or "HI" instead of a number

WHAT THE APP DOES TODAY

If the patient types "LO" or "HI", the app records no sugar reading and the check-in ends as **Routine**, in both versions of the interview. Nothing escalates.

WHY IT MATTERS

Home meters show "LO"/"HI" when sugar is below or above what the meter can measure (the exact limits vary by meter model).

YOUR DECISION

"LO" should be: ☐ Emergency ☐ Urgent ☐ Other: _____

"HI" should be: ☐ Emergency ☐ Urgent ☐ Other: _____

3 Low-sugar message that mentions "15 g" of sugar

WHAT THE APP SHOWS TODAY (IN THE NEWER INTERVIEW VERSION)

"Low blood sugar alert: If you feel shaky, sweaty, confused or dizzy, consume 15g fast-acting sugar (half cup juice or 3-4 glucose tablets) immediately."

کم شوگر کی اطلاع: اگر آپ کپکپاہٹ، پسینہ، الجھن، یا چکر محسوس کریں تو فوراً 15 گرام جلدی اثر کرنے والی شوگر (آدھا کپ رس یا 3 سے 4 گلوکوز کی گولیاں) لیں۔

Our project rule is that the app gives no dosing advice without doctor approval, so this needs your decision.

YOUR DECISION

- [ ] A. Keep it as it is (the standard "rule of 15")
- [ ] B. Replace with: *"Take quick sugar such as fruit juice or glucose, and seek immediate medical attention if symptoms persist."* (we'd translate it after your approval)
- [ ] C. Remove the amount and advise contacting a clinician immediately
- [ ] Other wording: _____

4 Blood pressure guideline edition

WHAT THE APP DOES TODAY

The app uses: Stage 2 hypertension from **140/90**, and a dangerous reading at **180/120**. Our documents cite the ACC/AHA **2017** guideline.

YOUR DECISION

- [ ] Cite the 2025 AHA/ACC guideline instead (thresholds unchanged)
- [ ] Keep citing 2017
- [ ] Change a threshold: _____

5 Non-stop vomiting / can't keep fluids down

WHERE WE ARE

The app treats this as **Urgent** today. Earlier reviews were split: Capt Dr Talha wrote *"nonstop vomiting is NOT emergency, it is an urgent review"*; Dr Shaiza agreed verbally; Dr Anam marked it **Emergency**. We'd like one shared decision.

YOUR DECISION

- [ ] (a) Urgent in all cases
- [ ] (b) Emergency in all cases
- [ ] (c) Urgent, but Emergency when it comes with: _____

6 Reminder to ask about an HbA1c test (people with high BP, no diabetes)

WHAT'S BUILT (SWITCHED OFF)

"Ask your doctor whether you need an HbA1c (sugar) test."

اپنے ڈاکٹر سے پوچھیں کہ کیا آپ کو HbA1c (شوگر) ٹیسٹ کی ضرورت ہے۔

It has no default interval; it stays off until you choose one.

YOUR DECISION

Remind if there's no HbA1c on record for: ☐ 12 months ☐ 24 months ☐ 36 months ☐ Keep it off

Wording: ☐ Approved ☐ Change to: _____

7 Doctor-only diabetes risk panel (experimental)

WHAT IT IS

For patients with high BP and no diabetes, the doctor's review screen can show an estimated chance of **undiagnosed** diabetes as **Low / Moderate / High** (no percentages), with the main contributing factors. Patients never see it. It is switched off and, even when on, it **cannot change** any triage level.

The model was trained on US survey data (NHANES). On US test data it ranks people correctly about 75% of the time (AUROC 0.75); in the group it's meant for (people with high BP) about 66% (AUROC 0.66). It has **not** been tested on Pakistani patients, and the panel says so.

YOUR DECISION

- ☐ Approve showing it to doctors only, as experimental, never changing a level
- ☐ Don't show it yet
- ☐ Comments: _____

8 Waist measurement: where to measure, and the allowed range

WHAT THE APP DOES TODAY

Accepts 40–200 cm. The help text says: *"Measure around your waist at the top of your hip bones, after breathing out."* (top of the hip bone, the iliac crest, which is also how the US data the model learned from was measured).

سانس باہر نکالنے کے بعد کولہے کی ہڈی کے اوپری حصے پر کمر کی پیمائش کریں۔

YOUR DECISION

Landmark: ☐ Top of the hip bone (current) ☐ WHO midpoint (between the lowest rib and the hip bone)

Range 40–200 cm: ☐ OK ☐ Change to: _____

9 Trend arrows (rising / falling) on home readings

PROPOSED

Show a rising/falling arrow over 30 days only if the change is at least **10 mmHg** (systolic), **5 mmHg** (diastolic) or **20 mg/dL** (sugar), and only with at least **5 readings**; otherwise "not enough readings yet". Arrows are information only; they never change a level.

YOUR DECISION

☐ Approve 10 / 5 / 20 and 5 readings ☐ Change: _____

10 A bare small number for sugar (2 to 40), unit unclear

WHAT THE APP DOES TODAY (THE VERSION PATIENTS USE)

If a patient answers the sugar question with a bare number between 2 and 40 and still doesn't give the unit when asked, the app treats it as a possible severe low: it saves it as mg/dL and starts the low-sugar path (Urgent, or Emergency if the patient is confused or can't eat or drink). For example, "9" might really mean 9 mmol/L (about 162 mg/dL, not low).

YOUR DECISION

- ☐ Keep it (safer to over-react)
- ☐ Ask once more; if still unclear, send to **Review** without saving a number
- ☐ Other: _____

11 A check-in with no readings at all

WHAT THE APP DOES TODAY

In the older interview version, a check-in with no sugar or BP reading goes to **Review**. In the newer version it ends as **Routine**. We need one rule before switching versions.

YOUR DECISION

- ☐ Always Review when there are no readings
- ☐ Routine if there are no symptoms; Review if any symptom is reported
- ☐ Other: _____

Summary and sign-off

#	Topic	Your choice (short)
1	Free-text danger signs (word groups)	
2	Meter "LO" / "HI"	
3	"15 g" low-sugar message	
4	BP guideline edition	
5	Non-stop vomiting	
6	HbA1c reminder interval	
7	Doctor-only risk panel	
8	Waist landmark and range	
9	Trend arrows	
10	Bare small sugar number	
11	No readings at all	

How you'd like to be named in our report (name, title, hospital), or "please don't name me":

_____

Name: _____ Signature: _____

Date: _____

Thank you. Every decision you make here is recorded with your name and date in our clinical rules document before anything in the app changes.